MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6072A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Twin triode, miniature, low noise

Outline --- 6-2 (EIA)

Base --- E9-1

Envelope --- T6-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	2a	2g	2k	h	h	1a	1g	1k	hct

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Pp/a	TE	Alt
Unit:	V	Vdc	Vdc	v	W	°C	ft
Maximum:	13.9	330	0	100	1.6	165	Note 1
	6.9	---	---	---	---	---	---
Minimum:	11.3	---	---	---	---	---	---
	5.7	---	---	---	---	---	---
TEST CONDITIONS:	12.6	250	-4	0	---	---	---

GENERAL:

Qualification - Required

Source: https://assist.dla.mil/ Downloaded: 2026-10-15T12:46Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1256	Electrode current (1) (anode)	See note 2	0.4	II	Ib	1.9	4.0	mAdc
1256	Electrode current (2) (anode)	Ec = -10.5 Vdc; Rp = 0.1 Meg (see note 2)	0.4	II	Ib	---	24	µAdc
1266	Total grid current	Rg = 0.25 Meg (see notes 2 and 3)	0.4	II	Ic	0	-0.5	µAdc
1301	Heater current		0.4	II	If	160	190	mA
1306	Transcoductance (1)	See note 2	0.4	II	Sm	1,350	2,150	µmhos
1336	Heater-cathode leakage	See note 4	0.4	II	Ihk	---	10	µAdc
1201	Short and discontinuity detection	See note 5	0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes	See note 2	2.5	I	R	2,000	---	Meg
1031	High-frequency vibration	Rp = 2,000 ohms (see note 4)	6.5	Code E	Ep	---	40	mVac
2201	Noise and microphonics	Ef = 6.3 Vdc; Ebb = 150 Vdc; Ecc = 0; Rg = 0.1 Meg; Rp = 20,000 ohms; Rk = 2,700 ohms; Ck = 40 µF; Ecal = 625 µVac (see notes 6 and 7)	2.5	II		(Reject for sustained howl)		
2201	Noise and microphonics (hiss)	Ef = 6.3 Vdc; Ebb = 150 Vdc; Ecc = 0; Rg = 0.1 Meg; Rp = 20,000 ohms; Rk = 2,700 ohms; Ck = 40 µF; Ecal = 200 µVac (see notes 6 and 7)	2.5	II	---	---	---	---
2201	Noise and microphonics (hum)	Ef = 6.3 Vac (60 cycles) Ebb = 150 Vdc; Ecc = 0; Rg = 0.1 Meg; Rp = 20,000 ohms; Rk = 2,700 ohms: Ecal = 200 µVac (see notes 6 and 8)	2.5	II	---	---	---	---
1256	Electrode current (1) (anode) (difference between sections)		2.5	I	ΔIb	---	0.9	mAdc
1266	Grid emission	Ef = 15 V; Ec = -10 Vdc; Rg/g = 0.25 Meg (see notes 2, 3, and 9)	2.5	I	Is	0	-1.5	µAdc
1306	Transconductance (2)	Ef = 11.4 V (see note 2)	2.5	I	ΔSm Ef	---	15	%
1316	Amplification Factor	See note 2	---	---	Mu	38	50	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:46Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
1321	Alternating current amplification	Ebb = 100 Vdc; Ecc = 0; Esig = 200 mVac; Rg = 10 Meg (see note 2)	2.5	I	Ep	5.5	---	Vac
1331	Direct-interelectrode capacitance	No shield (see note 2)	6.5	Code E	Cgp	1.1	1.7	pF
					Cin	1.2	1.8	pF
					Cout 1	0.24	0.72	pF
					Cout 2	0.19	0.57	pF
1121	Base strain		---	---	---	---	---	---
2126	Envelope strain		2.5	I	---	---	---	---
1041	Shock	630 G; Ecc = 0; Rg = 0.1 Meg; Rk = 1,200 ohms (see note 10)	6.5	See note 11	---	---	---	---
1031	Vibration-fatigue		6.5	See note 11	---	---	---	---
---	Shock and vibration-fatigue test end points:							
1031	High-frequency vibration		---	---	Ep	---	60	mVac
1336	Heater-cathode leakage		---	---	Ink	---	20	µAdc
1306	Transconductance (1)		---	---	Sm	1,200	---	µmhos
1266	Total grid current		---	---	Ic	0	-1.5	µAdc
1105	Permanence of marking		---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3					
1506	Heater-cycling life	Ef = 7.5 V; heaters in parallel; Enk = +135 Vdc; Ec = Eb = 0	---	---	---	---
---	Heater-cycling life-test end point:					
1336	Heater-cathode leakage		Ink	---	15	µAdc
1516	Stability life	Ecc = 0; Rg/g = 0.25 Meg; Rk/k = 375 ohms; TA = room (see note 12)	---	---	---	---
---	Stability life-test end point:					
1306	Change in transconductance (1) of individual tubes		ΔSm_t	---	15	%
1501	Intermittent life	Ecc = 0; Rg/g = 0.25 Meg; Rk/k = 375 ohms; TA = room; TE = +165°C (see notes 13 and 14)	---	---	---	---
---	Intermittent life-test end points (500 hours)	See note 15				
---	Inoperatives		---	---	---	---
1266	Total grid current		Ic	0	-0.5	µAdc
1301	Heater current		If	160	190	mA
1306	Change in transconductance (1) of individual tubes		ΔSm_t	---	20	%
1306	Transconductance (2)		ΔSm_{Ef}	---	15	%

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:46Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Quality conformance inspection, part 3 -Continued					
	Intermittent life-test end points (500 hours) -Continued					
1336	Heater-cathode leakage		Ink	---	10	µAdc
1211	Insulation of electrodes		R	1,000	---	Meg
2201	Noise and microphonics		Mic	(Reject for sustained howl)		
2201	Noise and microphonics (hiss)		Hiss	---	300	µVac
2201	Noise and microphonics (hum)		Hum	---	300	µVac
1306	Transconductance (1) average change		Avg ΔSm t	---	10	%
---	Total defectives		---	---	---	---
---	Intermittent life-test end points (1,000 hours):	See note 15				
---	Inoperatives		---	---	---	---
1266	Total grid current		Ic	0	-0.5	µAdc
1301	Heater current		If	160	190	mA
1306	Change in transconductance (1) of individual tubes		ΔSm t	---	25	%
1306	Transconductance (2)		ΔSm Ef	---	20	%
1336	Heater-cathode leakage		Ink	---	10	µAdc
1211	Insulation of electrodes		R	750	---	Meg
2201	Noise and microphonics		Mic	(Reject for sustained howl)		
2201	Noise and microphonics (hiss)		Hiss	---	300	µVac
2201	Noise and microphonics (hum)		Hum	---	300	µVac
---	Total defectives		---	---	---	---
1511	Cathode interface life	Ef = 6.9 V; heaters in parallel (see note 14)	Ri	---	25	ohms

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak in the basic document.
2. Test each unit separately.
3. This test shall be performed at the conclusion of the holding period.
4. Tie 1k to 2k; 1g to 2g; and 1a to 2a.
5. Tube shall not be tapped.
6. The TUT heaters shall be in parallel. The hum and noise level of the amplifier shall be read separately for each section with both sections operating and shall be at least 20 dB below the rejection level. The rejection level and the output level for calibration shall be 6 mW. Neither a mechanical tapper nor cork mallet tapping shall be used. Wire-wound noninductive resistors shall be used in cathode and anode circuits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:46Z
Check the source to verify that this is the current version before use.

NOTES: -Continued

7. The amplifier used in making this test shall possess the following characteristics:

   (a) A bandpass that extends from 40 to 13,040 Hz between the two, one-half power points.

   (b) A response within 1 dB of the response at 1,000 Hz over the frequency range of 60 to 7,500 Hz.

   (c) A cut-on and cutoff rate of 12 dB per octave.

   (d) Read hiss with the output load switch in the microphonics position.

   (e) Read microphonics with the output load switch in the noise position and tap tube under test lightly three times with the index finger (body grounded).

8. Tie the electrical center of the Ef supply to -B. The cathode resistor is not bypassed by a capacitor. The amplififer shall have a response within 0.5 dB of the response at 60 Hz over the frequency range of 50 to 150 Hz, with the upper half-power point at 400 ±40 Hz, and with a minimum cutoff rate (above 400 Hz) of 6 dB per octave. Read hum with the output load switch in the microphonics position.

9. Prior to this test, tubes shall be preheated a minimum of 5 minutes with both sections operating at the conditions specified below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef V	Ecc Vdc	Eb Vdc	Rk/k Ohms	Rg/g Meg
15.0	0	250	330	0.25

10. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is used.

11. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

12. Apply 2 Vdc from heater to ground (heater negative).

13. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

14. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.

15. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples.

16. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:46Z
Check the source to verify that this is the current version before use.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Air Force - 99
DLA - ES

User activities:
Navy - AS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3238)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:46Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Electronic Systems Command
Defense Standardization Program Branch
Department Of The Navy
Washington, D.C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:46Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-G-1/14E

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:46Z
Check the source to verify that this is the current version before use.